MIL-C-51515(EA)
28 May 1986
(See section 6)

MILITARY SPECIFICATION

CARTRIDGE, 5.56 MM: BLANK, M755

This specification is approved for use within Chemical Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a cartridge intended for propelling riot control grenades from the M234 launcher.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Standards. The following standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-414 - Sampling Procedures and Tables for Inspection by Variables for Percent Defective.

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, U.S. Army Chemical Research, Development and Engineering Center, ATTN: SMCCR-SPD-TS, Aberdeen Proving Ground, MD 21010-5423 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter. :

AMSC N/A FSC 1330

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MIL-STD-636 - Visual Inspection Standards for Small Arms Ammunition Through Caliber .50.
MIL-STD-644 - Visual Inspection Standards and Inspection Procedures for Inspection of Packaging, Packing and Marking of Small Arms Ammunition.
MIL-STD-1168 - Ammunition Lot Numbering.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS AND TECHNICAL DATA PACKAGE LIST (TDPL)

U.S. ARMY ARMAMENT, MUNITIONS AND CHEMICAL COMMAND

CHEMICAL RESEARCH AND DEVELOPMENT CENTER

TDPL 122-3-43 - Cartridge, 5.56 MM: Blank, M755.

FIRE CONTROL AND SMALL CALIBER WEAPONS SYSTEMS LABORATORY

8594163 - Seal, Car.
8595399 - Symbols, Boxes, Ammunition, M19A1 and M2A1.
10542289 - Carton, (5.56 MM, Grenade or Blank).
10542290 - Packing and Marking, Cartridges (Grenade or Blank) 5.56 MM; Cartons, Box, Ammunition, M2A1.
10542291 - Packing and Marking, Cartridges (Grenade or Blank) 5.56 MM; Cartons, Box, Ammunition, M2A1, Box, Wirebound.

U.S. ARMY MATERIEL DEVELOPMENT AND READINESS COMMAND

DARCOM Drawing 19-48-4116-20PA1002 - Unitization Procedures for Boxed Ammunition and Components on 4-Way Entry Pallets.
DARCOM Drawing 19-48-4116/5-20PA1002 - Unitization Procedures for Boxed Ammunition and Components on 4-Way Entry Pallets.

PUBLICATIONS

U.S. ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

Ammunition Ballistic Acceptance Test Methods, Test Procedures for 5.56MM Cartridges (SCATP 5.56)

(Copies of standards, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

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2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on TDPL 122-3-43 and on the subsidiary drawings shall conform to the specifications listed thereon, or to the specific characteristics set forth on the drawings.

3.1.2 Components. All components of the cartridge shall conform to the specifications and drawings listed on TDPL 122-3-43.

3.2 Assembly. The cartridge shall be assembled as shown on Drawing 122-3-43.

3.3 Residual stress. The blank cartridge case shall not split when subjected to a one percent mercurous nitrate solution for 15 minutes when tested in accordance with 4.5.3.

3.4 Functioning. The cartridge shall function when tested in accordance with 4.5.1.

3.5 Filling. The cartridge shall contain 12.15 $\pm$ 0.30 grains of powder when tested in accordance with 4.5.2.

3.6 Preproduction. Prior to the start of regular production, a preproduction lot of cartridges shall be produced in accordance with this specification for examination and tests (see 4.3).

3.7 Waterproof. The blank cartridge shall not release more than one bubble of air when subjected to a pressure of 2 pounds per square inch (psi) below atmospheric pressure and held at that pressure for 15 seconds.

3.8 Workmanship. The requirements for workmanship are as specified on the applicable drawings, referenced specification, and the following:

3.8.1 Metal defects. The cartridge shall be free of folds, wrinkles, deep draw scratches, scaly metal, dents and other defects.

3.8.2 Foreign matter. The cartridge shall be free of corrosion, stains, discolorations, dirt, oil and smears of lacquer.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Preproduction inspection.

4.3.1 Lot. A preproduction lot of 2280 cartridges, packaged and packed (6.3), shall be manufactured using the same methods, materials, equipment, and processes as will be used during regular production.

4.3.2 Inspection procedure.

4.3.2.1 For examination. A sample of 10 cartridges, 5 packaging cartons, 2 intermediate packaging boxes and 1 packing box (6.3) shall be examined to all drawing dimensions. The acceptance number is zero.

4.3.2.2 For tests.

(a) Functioning. A sample of 100 cartridges shall be tested in accordance with 4.5.1. The acceptance number is zero.

(b) Residual stress. A sample of 50 cartridges shall be tested in accordance with 4.5.3. The acceptance number is zero.

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(c) Filling weight. A sample of 40 cartridges shall be tested in accordance with MIL-STD-414 and 4.5.2 using an AQL of 1.0 percent defective, tightened inspection.

(d) Waterproof. A sample of 20 cartridges shall be tested in accordance with 4.5.4. Failure of seven or more cartridges to comply with the applicable requirement shall reject the lot. If more than two, but less than seven cartridges fail in the first test, a second sample consisting of 20 cartridges shall be tested. The lot shall be rejected if, in the combined first and second sample, seven or more cartridges fail to comply with the applicable requirements.

4.3.3 Acceptance/rejection criteria. The preproduction sample shall comply with the examinations and tests specified in 4.3.2 to be acceptable. The contractor shall obtain written approval from the contracting activity prior to [illegible] production.

4.4 Quality conformance inspection.

4.4.1 Lotting. Each lot shall be identified and controlled in accordance with MIL-STD-1168. However, no more than one batch of powder (see 6.2) from one supplier shall be represented in any one lot of assembled cartridges.

4.4.2 Sampling.

4.4.2.1 For examination. Sampling shall be conducted in accordance with MIL-STD-105.

4.4.2.2 For destructive tests.

(a) Functioning. Sampling shall be conducted in accordance with MIL-STD-105, level S-4.

(b) Filling weight. Sampling shall be conducted in accordance with MIL-STD-414, level IV.

(c) Residual stress. A random sample of 50 each shall be tested per lot.

(d) Waterproof. A random sample of 20 each shall be tested per lot.

4.4.3 Inspection procedure.

4.4.3.1 For examination. Examination for major and minor defects shall be performed on a class basis in accordance with the classification of defects, table I, using applicable sampling plans and acceptance criteria of MIL-STD-105. The Acceptance Quality Level (AQL) for the major class shall be 0.65 percent and the AQL for the minor class shall be 2.50 percent.

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4.4.3.2 For destructive tests.

(a) Functioning. The sample cartridges shall be tested in accordance with 4.5.1 and with MIL-STD-105 using an AQL of 0.40 percent defective.

(b) Filling weight. The sample cartridges shall be tested in accordance with 4.5.2 and with MIL-STD-414 using an AQL of 1.0 percent defective.

(c) Residual stress. The sample cartridges shall be tested in accordance with 4.5.3. Failure of two or more cartridges to comply with the applicable requirement shall be cause for rejection of the lot. If one cartridge fails in the first test, a second sample consisting of double the number of cartridges in the first sample shall be tested. If any failing cartridges are found in the second sample, the lot shall be rejected.

(d) Waterproof. The sample cartridges shall be tested in accordance with 4.5.4. Failure of nine or more cartridges to comply with the applicable requirement shall be cause for rejection of the lot. If more than four, but less than nine cartridges fail in the first test, a second sample consisting of double the number of cartridges in the first sample shall be tested. The lot shall be rejected if, in the combined first and second sample, nine or more cartridges fail to comply with the applicable requirement.

4.4.3.3 Classification of defects. The classification of defects shall be as specified in table I.

4.4.4 Packaging, packing and marking inspection. During or immediately prior to the packaging operation, 100 percent examination of the cartridges shall be performed to ascertain that the cartridge type conforms to the drawing. Occurrence of another model blank cartridge or a dummy cartridge shall be classed as a major defect. Occurence of any type other than those listed shall be classed as a critical defect. All nonconforming cartridges shall be rejected. Inspection for packaging, packing and marking shall be in accordance with MIL-STD-644 as applicable to the drawing.

4.5 Tests.

4.5.1 Functioning. The cartridges shall be immersed in water at ambient temperature to a depth of 2.0 ± 0.5 inch for a period of 30 (+5, -0) minutes. At the end of the immersion period, the cartridge shall be dried and functioned within 1 hour in a fixture acceptable to the U.S. Government, the AR15 or M16 rifle. The cartridge shall fire, the primer shall not blow or fall out of the pocket upon retraction of the bolt. The cartridge case body and head shall not split or rupture and no particle(s) of metal shall become separated from the case between the mouth and the juncture of the shoulder and neck.

4.5.2 Filling weight. The powder shall be removed from the cartridge and weighed to the nearest .01 grain.

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TABLE I. Classification of defects

No.	Defect and Method of Inspection	Major	Minor
	Visual[1/]		
1.	Discolored, dirty, oily, smeared		X
2.	Corroded or stained, if etched		X
4.	Round head	X	
5.	Dent		X
6.	Split [2/]	X	
7.	Perforation	X	
8.	Draw scratch		X
10.	Beveled underside of head	X	
12.	Scaly metal		X
13.	No chamfer on head (rim)		X
14.	Fold		X
18.	Illegible or missing head stamp		X
19.	Defective head		X
32.	No primer	X	
33.	Cocked primer	X	
34.	Inverted primer	X	
35.	Loose primer	X	
36.	Nicked or dented primer		X
37.	No waterproofing material (primer pocket joint)		X
38.	Defective primer crimp		X
40.	Mouth waterproofing missing	X	
41.	Mouth crimp missing or incorrect	X	
	Gaging		
42.	Total length		X
43.	Cartridge profile failure (requiring more than 10 pounds dead weight to insert in profile and alignment gage, Drawing 8654100)	X	
44.	Diameter of extractor groove, max.	X	
45.	Diameter of extractor groove, min.		X
46.	Diameter of head, max.	X	
47.	Diameter of head, min.		X
48.	Thickness of head		X
49.	Length of shoulder datum (Drawing 8654101)	X	
50.	Depth of primer	X	

[1/] Refer to MIL-STD-636 for visual standards for defects 1 through 38. While no photographs of blank cartridges are included in this Standard, the cartridge case defects illustrated in the NATO 7.62mm section can be used. In the event of conflict between table I of this specification and MIL-STD-636 as to defect classification, the classification specified in table I shall apply.

[2/] A split shall be classed as a major defect regardless of location or occurrence of loss of propellant.

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4.5.3 Residual stress. Test shall be conducted in accordance with appendix.

4.5.4 Waterproof. The test shall be conducted in accordance with SCATP-5.56.

5. PACKAGING

5.1 Packaging, level A. Twenty cartridges shall be packaged together in accordance with Drawing 10542289.

5.2 Intermediate packaging, level A. Cartridges packaged as specified in 5.1 shall be intermediately packaged together in accordance with Drawing 10542290.

5.3 Packing, level A. Cartridges intermediately packaged as specified in 5.2 shall be packed as specified on Drawings 10542291 and 8594163.

5.4 Marking. Cartridges packaged as specified in 5.1 shall be marked in accordance with the marking instructions cited on Drawing 10542289. Cartridges intermediately packaged as specified in 5.2 shall be marked in accordance with the marking instructions cited on Drawings 10542290 and 8595399. Cartridges packed as specified in 5.3 shall be marked in accordance with the marking instructions cited on Drawing 10542291.

5.5 Palletization. When specified in the contract or order, items packed in accordance with 5.3 shall be palletized in accordance with Drawings 19-48-4116-20PA1002 and 19-48-4116/5-20PA1002.

6. NOTES

6.1 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Level A packaging required.
(c) Preproduction:

(1) Time allowed for contractor submission of samples for Government test and evaluation after award of contract.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify the contractor whether or not to proceed with production.

6.2 Batch. A batch is defined as that quantity of material which has been manufactured by some chemical process or subjected to some physical mixing operation to make the final product substantially uniform.

6.3 Caution. The primed blank cartridge cases are classified Class "C" explosives with minimum of susceptibility to static discharge detonation. They should be handled in a careful manner with caution against impact and shock

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loading. The propellant utilized is classified Class "B" propellant and is flammable and will burn in the absence of oxygen since it contains its own oxidant.

6.4 Supersession data. This specification supersedes and includes the requirements of Chemical Systems Laboratory Purchase Description EA-C-1144B, dated 28 June 1978.

Custodian:

Army - EA

Preparing activity:

Army - EA

Project No. 1330-A236

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APPENDIX

TEST PROCEDURE FOR

RESIDUAL STRESS (MERCUROUS NITRATE)

10. SCOPE

10.1 Scope. This appendix details a test for visual means of determining if stresses exist in brass cartridge cases that may cause the cases to split under conditions of storage or during service usage.

20. MANDATORY SAFETY REQUIREMENTS

20.1 Food shall not be stored or eaten in vicinity in which these tests are conducted.

20.2 Acid resistant apron and gloves or the equivalent shall be worn by each test technician.

20.3 Face shields shall be worn at all times during the pouring or mixing of acids and water. Safety glasses shall be worn during other phases of this test.

20.4 Asbestos or heat insulating gloves shall be worn during the heat volatilization phase of the test to facilitate handling.

20.5 During the entire period of volatilization the oven door shall be closed.

20.6 Extreme care shall be exercised in the mixing of acid with water; this shall be accomplished by pouring the acid into the water.

20.7 Test shall be conducted under a canopy or hood having a forced draft ventilation system to remove the noxious fumes. The disposition of tested components shall be governed by local regulations.

30. TEST PROCEDURES

30.1 Preparation of solutions.

30.1.1 Nitric-acid solution. Four hundred (400) milliliters of nitric acid (of specific gravity 1.42) are dissolved in five hundred (500) milliliters of distilled water at room temperature. To this solution, distilled water is added to bring the volume of the resulting solution to one liter. (The resultant specific gravity will be 1.25.)

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30.1.2 Mercurous nitrate solution. Ten (10) grams of mercurous nitrate and ten (10) milliliters of nitric acid (of specific gravity 1.42) are dissolved in four hundred (400) milliliters of distilled water at room temperature. To this solution, distilled water is added to bring the volume of the resulting solution to one liter.

30.2 Test of cartridge cases.

30.2.1 The cartridges are positioned in both nitric acid and mercurous nitrate solutions in a vertical position, with the head of the case down. The depth of each solution is adjusted until it completely covers the mouth of the case.

30.2.2 The test cartridges are submerged in the nitric acid solution. After thirty (30) seconds the cartridges are withdrawn, rinsed in water and the excess water removed. The cartridges are then submerged for fifteen (15) minutes in the mercurous nitrate solution. Upon removal, the surface of the cartridge case is examined under a magnification (10 to 15 diameters) for splits and cracks. All splits and cracks found are reported.

NOTE: A split is defined as a separation of the metal entirely through the wall of the cartridge case. A crack is a surface condition and represents a separation of the metal partially through the wall of the cartridge case. Cracks are not considered to be splits.

40. RECORDING OF DATA

40.1 All splits shall be reported.

40.2 All cracks shall be reported for information only.

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NOTICE OF INACTIVATION FOR NEW DESIGN

INCH-POUND

MIL-C-51515(EA)
NOTICE 1
8 November 1995

MILITARY SPECIFICATION

CARTRIDGE, 5.56 MM: BLANK, M755

This notice should be filed in front of MIL-C-51515(EA) dated 28 May 1986.

MIL-C-51515(EA) is inactive for new design and is no longer used except for replacement purposes.

Preparing activity:

Army - EA

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